

Task 1: To Create Pipes, Redirect Streams, Execute Connected Commands, Close Descriptors, Synchronize Processes, Test Data Flow

```
GNU nano 7.2 task1.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    int pipefd[2];

    if (pipe(pipefd) == -1) {
        perror("pipe");
        return 1;
    }

    pid_t pid = fork();

    if (pid < 0) {
        perror("fork");
        return 1;
    }

    if (pid == 0) {
        close(pipefd[1]);

        dup2(pipefd[0], STDIN_FILENO);
        close(pipefd[0]);

        if (pid == 0) {
            close(pipefd[1]);

            dup2(pipefd[0], STDIN_FILENO);
            close(pipefd[0]);

            execlp("sort", "sort", NULL);

            perror("execlp");
            exit(1);
        }

        close(pipefd[0]);

        dup2(pipefd[1], STDOUT_FILENO);
        close(pipefd[1]);

        execlp("ls", "ls", NULL);

        perror("execlp");
        wait(NULL);

        return 1;
    }
}
```

```

shreyas@Grigori:~$ mkdir -p ~/pipe_tasks
shreyas@Grigori:~$ cd ~/pipe_tasks
shreyas@Grigori:~/pipe_tasks$ nano task1.c
shreyas@Grigori:~/pipe_tasks$ gcc task1.c -o task1
shreyas@Grigori:~/pipe_tasks$ ./task1
task1
task1.c
shreyas@Grigori:~/pipe_tasks$ ./task1 > output.txt
shreyas@Grigori:~/pipe_tasks$ cat output.txt
output.txt
task1
task1.c

```

Task 2: To Support Multiple Pipes, Manage Descriptor Arrays, Launch Multiple Processes, Coordinate Execution, Cleanup Resources, Test Long Pipelines

```

GNU nano 7.2 task2.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

#define N 4

int main() {
    int pipes[N - 1][2];
    pid_t pids[N];

    for (int i = 0; i < N - 1; i++) {
        if (pipe(pipes[i]) == -1) {
            perror("pipe");
            return 1;
        }
    }

    char *commands[N][4] = {
        {"ls", NULL},
        {"sort", NULL},
        {"uniq", NULL},
        {"wc", "-l", NULL}
    };

    for (int i = 0; i < N; i++) {

```

```

pids[i] = fork();

if (pids[i] < 0) {
    perror("fork");
    return 1;
}

if (pids[i] == 0) {
    if (i > 0) {
        dup2(pipes[i - 1][0], STDIN_FILENO);
    }

    if (i < N - 1) {
        dup2(pipes[i][1], STDOUT_FILENO);
    }

    for (int j = 0; j < N - 1; j++) {
        close(pipes[j][0]);
        close(pipes[j][1]);
    }

    execvp(commands[i][0], commands[i]);
}

```

```

    }

    execvp(commands[i][0], commands[i]);

    perror("execvp");
    exit(1);
}

for (int i = 0; i < N - 1; i++) {
    close(pipes[i][0]);
    close(pipes[i][1]);
}

for (int i = 0; i < N; i++) {
    waitpid(pids[i], NULL, 0);
}

return 0;
}

```

```

shreyas@Grigori:~/pipe_tasks$ nano task2.c
shreyas@Grigori:~/pipe_tasks$ gcc task2.c -o task2
shreyas@Grigori:~/pipe_tasks$ ./task2
5
shreyas@Grigori:~/pipe_tasks$ ls | sort | uniq | wc -l
5
shreyas@Grigori:~/pipe_tasks$ ps
  PID TTY          TIME CMD
  618 pts/2    00:00:00 bash
  670 pts/2    00:00:00 ps
shreyas@Grigori:~/pipe_tasks$ ls | sort | uniq | wc -l
5
shreyas@Grigori:~/pipe_tasks$ lsof -p $$
COMMAND PID  USER  FD  TYPE DEVICE SIZE/OFF  NODE NAME
bash    618  shreyas  cwd  DIR   8,48  4096 37774 /home/shreyas/pipe_tasks
bash    618  shreyas  rtd  DIR   8,48  4096 2 /
bash    618  shreyas  txt  REG   8,48 1446024 1465 /usr/bin/bash
bash    618  shreyas  mem  REG   8,48 2129424 49240 /usr/lib/x86_64-linux-gnu/libc.so.6
bash    618  shreyas  mem  REG   8,48 360460 16512 /usr/lib/locale/C.utf8/LC_CTYPE
bash    618  shreyas  mem  REG   8,48 50 16518 /usr/lib/locale/C.utf8/LC_NUMERIC
bash    618  shreyas  mem  REG   8,48 3360 16521 /usr/lib/locale/C.utf8/LC_TIME
bash    618  shreyas  mem  REG   8,48 1406 16511 /usr/lib/locale/C.utf8/LC_COLLATE
bash    618  shreyas  mem  REG   8,48 270 16516 /usr/lib/locale/C.utf8/LC_MONETARY
bash    618  shreyas  mem  REG   8,48 48 16515 /usr/lib/locale/C.utf8/LC_MESSAGES/SYS_LC_MESSAGES
bash    618  shreyas  mem  REG   8,48 27028 48959 /usr/lib/x86_64-linux-gnu/gconv/gconv-modules.cache
bash    618  shreyas  mem  REG   8,48 208328 12112 /usr/lib/x86_64-linux-gnu/libtinfo.so.6.4
bash    618  shreyas  mem  REG   8,48 34 16519 /usr/lib/locale/C.utf8/LC_PAPER
bash    618  shreyas  mem  REG   8,48 62 16517 /usr/lib/locale/C.utf8/LC_NAME
bash    618  shreyas  mem  REG   8,48 127 16510 /usr/lib/locale/C.utf8/LC_ADDRESS

```

```

bash    618  shreyas    0u  CHR 136,2    0t0    5 /dev/pts/2
bash    618  shreyas    1u  CHR 136,2    0t0    5 /dev/pts/2
bash    618  shreyas    2u  CHR 136,2    0t0    5 /dev/pts/2
bash    618  shreyas    7u  CHR 5,2      0t0   86 /dev/ptmx
bash    618  shreyas   255u  CHR 136,2    0t0    5 /dev/pts/2
shreyas@Grigori:~/pipe_tasks$ rm -f task1 task2 output.txt
shreyas@Grigori:~/pipe_tasks$ ls
task1.c task2.c
shreyas@Grigori:~/pipe_tasks$ |

```